Experiment 8

Ex. No.: 8

Collecting Email Evidence in Victim PC

Aim

To understand an authorised Windows memory-acquisition workflow for locating email-address artefacts with DumpIt and Bulk Extractor Viewer.

Tools required

1. Authorised Windows lab machine.
2. DumpIt memory-acquisition utility.
3. Bulk Extractor and Bulk Extractor Viewer.
4. Sufficient external storage for the raw memory image and extracted report.

Idea/Theory

1. Volatile memory can contain email addresses and fragments of application or message data that are not present in ordinary files.
2. DumpIt creates a physical memory image of supported 32-bit and 64-bit Windows systems after confirmation.
3. Bulk Extractor scans an image for features and produces files such as `email.txt` and `email_histogram.txt`.
4. Live acquisition changes the running system; the examiner must record the authority, tool version, destination, time context, and hash.

Experiment 8 — continued

Procedure/Steps

1. Prepare an authorised lab system and clean destination media with enough free space.
2. Run `DumpIt.exe` from the approved location.
3. Read the displayed source and destination details, then enter Y only after they are confirmed.
4. Allow DumpIt to write the RAW memory image to the selected directory.
5. Record the image size and calculate a cryptographic hash before analysis.
6. Install and open Bulk Extractor Viewer, then choose the option to generate a report.

Experiment 8 — continued

Procedure/Steps (continued)

1. Select the DumpIt RAW image as the input.
2. Choose a separate output folder and start Bulk Extractor.
3. After processing, open `email.txt` to review extracted email-address strings.
4. Open `email_histogram.txt` to review counts and prioritise repeated entries.
5. Validate any relevant string in context; do not treat an extracted address alone as proof of authorship or communication.

Observation/Output

Screenshots to be attached from the lab run (tool: DumpIt and Bulk Extractor Viewer)

Result

To be completed after the lab run.